



9292 - How Large Stellar Flares Influence the Chemistry of Protoplanetary Disks

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				

Folder	Observation	Label	Observing Template	Science Target
	1	W1P1_Orion_Nebula	NIRSpec MultiObject Spectroscopy	(2) OrionNebula_v2
	2	W1P2_Orion_Nebula	NIRSpec MultiObject Spectroscopy	(2) OrionNebula_v2
	3	W1P3_Orion_Nebula	NIRSpec MultiObject Spectroscopy	(2) OrionNebula_v2
	4	W1P4_Orion_Nebula	NIRSpec MultiObject Spectroscopy	(2) OrionNebula_v2
	5	W2P1_Orion_Nebula	NIRSpec MultiObject Spectroscopy	(2) OrionNebula_v2
	6	W2P2_Orion_Nebula	NIRSpec MultiObject Spectroscopy	(2) OrionNebula_v2
	7	W2P3_Orion_Nebula	NIRSpec MultiObject Spectroscopy	(2) OrionNebula_v2
	8	W2P4_Orion_Nebula	NIRSpec MultiObject Spectroscopy	(2) OrionNebula_v2

ABSTRACT

Empirical understanding of flare-driven chemistry in protoplanetary disks is lacking, despite recent theoretical advancements. We propose near-simultaneous 70 ks Chandra/ACIS-I and ~40 hr JWST/NIRSpec-MSA observations of the Orion Nebula region to address this gap. For a large sample of disk-bearing young stars, Chandra will identify subsets of X-ray non-flaring stars and stars experiencing large flares, while JWST will obtain their infrared spectra. By comparing mass-stratified disk properties across these subsets and testing them against thermochemical models, we will gain critical insights into disk evolution and planet formation. Large-scale studies of flare-driven disk chemistry, spanning extensive samples of both flaring and non-flaring stars, have never been conducted before. For reference, the Chandra proposal ID for this joint Chandra/JWST program is 27200169.

OBSERVING DESCRIPTION

Preliminary theoretical tests using the ProDiMo (Protoplanetary Disk Model) thermochemical code suggest that X-ray/extreme-UV photons and relativistic suprathermal particles from large stellar flares can ionize hydrogen and heavier elements in surrounding protoplanetary disks, driving chemical changes that enhance the spectral line fluxes of HI, OH, and CO by 100% or more (Figure 2 in the proposal). We propose a sample of 150 disk-bearing stars in the Orion Nebula region (all with Gaia DR3 counterparts) to empirically test these theoretical predictions (Figure 3).

Over a ~114-hour observing window (Figure 4), six Chandra snapshots will identify X-ray flaring and non-flaring stars, followed by eight nearly-simultaneous JWST NIRSpec-MSA snapshots (four MSA pointings, each repeated twice) to obtain infrared spectra.

The Orion Nebula has a favorable JWST and Chandra feasibility window from January 20 to March 18, 2026. The observing dates (orange tick marks in Figure 4) are currently fixed in Chandra CPS and JWST APT, starting February 1, 2026, at 00:00:00 UT. However, both planning teams

JWST Proposal 9292 (Created: Monday, December 1, 2025, 5:01:19PM Eastern Standard Time) - Overview

may shift the timeline within the January-March feasibility window to resolve scheduling conflicts. MSA pointing setups were tested for APA angles of 188-232 degrees (January-March, 2026), achieving an 82% success rate in capturing spectra for >120 of the 150 proposed stars. We anticipate that Chandra will find ~30 of these stars in an X-ray flaring state.

NIRSpec spectra will be obtained in G395H/F290LP high-resolution mode ($R=2700$) over the 2.87-5.14 micron range to resolve numerous HI, OH, and CO lines. For simulated non-flaring spectra of 0.7 Mo (Figure 2) and 0.2 Mo (not shown) stars, JWST ETC predicts that CO lines in the >4.3 micron band will be detectable at >25-sigma significance, with a continuum SNR>500 for the 0.7 Mo star and >15-sigma (SNR>350) for the 0.2 Mo star --- more than sufficient to detect $\geq 100\%$ flare-driven spectral enhancements. Many OH and HI lines are expected to be detectable only in the flaring state. To achieve this, we request 4 groups (to avoid saturation), 10 integrations, and the NRSRAPID readout pattern for each of the three nod positions in each MSA configuration (out of five MSA configurations for each of the 4 proposed MSA pointings; Figure 3a in the proposal). Some massive stars with bright disks will be partially saturated below 4 micron, but their spectra beyond 4 micron will remain unsaturated and scientifically useful (Figure 3b).

NIRCam pre-imaging is unnecessary since existing NIRCam images (Luhman 2024a, AJ, 168, 230) fully cover three MSA pointings, and HST WFC3 images (Robberto et al. 2020, ApJ, 896, 79) cover all four MSA pointings. We estimate 20-50 faint target acquisition (TA) candidates per pointing ($F_{140X} = 20.6-23$ mag), already cataloged in Robberto et al. and Luhman et al. The full Orion Nebula NIRCam source catalog is also available to us (private communication). HST WFC3 TA candidates were successfully used in the NIRSpec-MSA study of Orion brown dwarfs, even without NIRCam data (Luhman et al. 2024b, ApJ, 975, 162). All proposed JWST point sources (150 disk-bearing stars and TA candidates) will be provided in the Gaia DR3 astrometric reference frame.

Proposal 9292 - Targets - How Large Stellar Flares Influence the Chemistry of Protoplanetary Disks

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	Orion_Nebula	RA: 05 35 18.0954 (83.8253975d) Dec: -05 23 9.34 (-5.38593d) Equinox: J2000		
	Comments: The current sample of 150 disk young stars does not include fainter TA candidates. A comprehensive list of all known stellar objects, including numerous TA candidates across our four MSA pointings, will be provided at a later time. This list is derived from existing JWST NIRCcam data (Luhman 2024, AJ, 168, 230) and HST WFC3 mosaics (Robberto et al. 2020, ApJ, 896, 79). Description=[]				
Fixed Targets	(2)	OrionNebula_v2	RA: 05 35 17.1187 (83.8213279d) Dec: -05 22 49.41 (-5.38039d) Equinox: J2000		
	Comments: Description=[]				

Proposal 9292 - Observation 1 - How Large Stellar Flares Influence the Chemistry of Protoplanetary Disks

Observation	Proposal 9292, Observation 1: W1P1_Orion_Nebula											Mon Dec 01 22:01:19 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Comments: The proposed Chandra/JWST timeline is depicted in Figure 4 of the proposal. All observing dates (orange tick marks) are currently fixed in the Chandra CPS and JWST APT forms, starting from February 1, 2026, 00:00:00 UT. However, the Chandra and JWST planning teams may shift this timeline within the January-March, 2026 feasibility window to resolve any proposal date conflicts.											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(2)	OrionNebula_v2	RA: 05 35 17.1187 (83.8213279d) Dec: -05 22 49.41 (-5.38039d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	Filter: F140X; Readout: NRSRAPID; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold	
	MSATA	false		No	MSA Center	PrimaryNew_v2 (150 sources)			jwst-nirspec-g395h		1.5	
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude		
	1	155	83.843136	-5.316797	21.4	1	162	83.836585	-5.280693	21.6		
	1	156	83.842111	-5.316007	21.0	1	163	83.839405	-5.286343	21.4		
	1	157	83.822241	-5.309248	21.8	1	165	83.810248	-5.293962	21.3		
	1	160	83.845795	-5.283576	21.3	1	171	83.799990	-5.275449	21.6		

Proposal 9292 - Observation 1 - How Large Stellar Flares Influence the Chemistry of Protoplanetary Disks

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G395H/F290LP)	c1 : W1P1obs1a	3 Shutter Slitlet	83.823670958333 32 Degrees - 5.2913674999999 785 Degrees	202.57570940441 101			3	30	1610.515
	2	1 (G395H/F290LP)	c1 : W1P1obs1b	3 Shutter Slitlet	83.818269541666 67 Degrees - 5.2896880555555 3 Degrees	202.57620520363 898			3	30	1610.515
	3	1 (G395H/F290LP)	c1 : W1P1obs1c	3 Shutter Slitlet	83.828184833333 34 Degrees - 5.2954624999999 85 Degrees	202.57530172413 973			3	30	1610.515
	4	1 (G395H/F290LP)	c1 : W1P1obs1d	3 Shutter Slitlet	83.829021375 Degrees - 5.2866936111111 045 Degrees	202.57520279072 26			3	30	1610.515
	5	1 (G395H/F290LP)	c1 : W1P1obs1e	3 Shutter Slitlet	83.822658208333 34 Degrees - 5.2886919444444 52 Degrees	202.57579649047 807			3	30	1610.515
Special Requirements	Between Dates 01-FEB-2026:09:00:00 and 01-FEB-2026:10:06:00 MSA Scheduled Aperture PA 202.5761 to 202.5761 Degrees (V3 64.00155 to 64.00155)										

Proposal 9292 - Observation 2 - How Large Stellar Flares Influence the Chemistry of Protoplanetary Disks

Observation	Proposal 9292, Observation 2: W1P2_Orion_Nebula										Mon Dec 01 22:01:19 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRSpect MultiObject Spectroscopy										
	Comments: The proposed Chandra/JWST timeline is depicted in Figure 4 of the proposal. All observing dates (orange tick marks) are currently fixed in the Chandra CPS and JWST APT forms, starting from February 1, 2026, 00:00:00 UT. However, the Chandra and JWST planning teams may shift this timeline within the January-March, 2026 feasibility window to resolve any proposal date conflicts.										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous	
	(2)	OrionNebula_v2	RA: 05 35 17.1187 (83.8213279d) Dec: -05 22 49.41 (-5.38039d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	Filter: F140X; Readout: NRSRAPID; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold
	MSATA	false		No	MSA Center	PrimaryNew_v2 (150 sources)		jwst-nirspect-g395h		1.5	
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	273	83.845537	-5.345231	21.1	1	295	83.803747	-5.373326	21.9	
	1	282	83.829511	-5.341369	21.1	1	488	83.805184	-5.369404	21.6	
	1	288	83.820383	-5.379229	21.3	1	490	83.804377	-5.359514	22.2	
	1	291	83.811259	-5.338059	21.1	1	508	83.795054	-5.363559	22.2	

Proposal 9292 - Observation 2 - How Large Stellar Flares Influence the Chemistry of Protoplanetary Disks

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G395H/F290LP)	c1 : W1P2obs2a	3 Shutter Slitlet	83.816957208333 33 Degrees - 5.35786111111111 06 Degrees	202.47430752059 11			3	30	1610.515
	2	1 (G395H/F290LP)	c1 : W1P2obs2b	3 Shutter Slitlet	83.821026333333 34 Degrees - 5.35747777777777 74 Degrees	202.47392913827 073			3	30	1610.515
	3	1 (G395H/F290LP)	c1 : W1P2obs2c	3 Shutter Slitlet	83.812450625 Degrees - 5.34788861111111 02 Degrees	202.47470506450 94			3	30	1610.515
	4	1 (G395H/F290LP)	c1 : W1P2obs2d	3 Shutter Slitlet	83.8250595 Degrees - 5.34893333333333 21 Degrees	202.47353755388 562			3	30	1610.515
	5	1 (G395H/F290LP)	c1 : W1P2obs2e	3 Shutter Slitlet	83.820530124999 99 Degrees - 5.35979861111111 03 Degrees	202.47397986183 4			3	30	1610.515
Special Requirements	Between Dates 01-FEB-2026:18:00:00 and 01-FEB-2026:19:06:00 MSA Scheduled Aperture PA 202.4739 to 202.4739 Degrees (V3 63.899376 to 63.899376)										

Proposal 9292 - Observation 3 - How Large Stellar Flares Influence the Chemistry of Protoplanetary Disks

Observation	Proposal 9292, Observation 3: W1P3_Orion_Nebula											Mon Dec 01 22:01:19 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(2)	OrionNebula_v2	RA: 05 35 17.1187 (83.8213279d) Dec: -05 22 49.41 (-5.38039d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	Filter: F140X; Readout: NRSRAPID; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold				
	MSATA	false	No	MSA Center	PrimaryNew_v2 (150 sources)		jwst-nirspec-g395h	1.5				
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude		
	1	229	83.873397	-5.390865	21.5	1	270	83.850944	-5.436598	21.1		
	1	235	83.869966	-5.388101	21.9	1	290	83.817643	-5.415124	21.2		
	1	253	83.863076	-5.430533	21.2	1	457	83.841206	-5.390093	21.9		
	1	269	83.847776	-5.391330	21.9	1	463	83.833459	-5.422882	21.1		

Proposal 9292 - Observation 3 - How Large Stellar Flares Influence the Chemistry of Protoplanetary Disks

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G395H/F290LP)	c1 : W1P3obs3a	3 Shutter Slitlet	83.853358333333 33 Degrees - 5.40667611111110 96 Degrees	203.20113241069 708			3	30	1610.515
	2	1 (G395H/F290LP)	c1 : W1P3obs3b	3 Shutter Slitlet	83.8496875 Degrees - 5.4047316666666 92 Degrees	203.20146978853 563			3	30	1610.515
	3	1 (G395H/F290LP)	c1 : W1P3obs3c	3 Shutter Slitlet	83.8605485 Degrees - 5.4049241666666 6 Degrees	203.20046292172 364			3	30	1610.515
	4	1 (G395H/F290LP)	c1 : W1P3obs3d	3 Shutter Slitlet	83.852616125 Degrees - 5.4095627777777 51 Degrees	203.20120584114 85			3	30	1610.515
	5	1 (G395H/F290LP)	c1 : W1P3obs3e	3 Shutter Slitlet	83.856168375 Degrees - 5.40663611111110 84 Degrees	203.20087173970 2			3	30	1610.515
Special Requirements	Between Dates 02-FEB-2026:03:00:00 and 02-FEB-2026:04:06:00 MSA Scheduled Aperture PA 203.2041 to 203.2041 Degrees (V3 64.62949 to 64.62949)										

Proposal 9292 - Observation 4 - How Large Stellar Flares Influence the Chemistry of Protoplanetary Disks

Observation	Proposal 9292, Observation 4: W1P4_Orion_Nebula										Mon Dec 01 22:01:19 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	OrionNebula_v2	RA: 05 35 17.1187 (83.8213279d) Dec: -05 22 49.41 (-5.38039d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	Filter: F140X; Readout: NRSRAPIDD6; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153	
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold
	MSATA	false		No	MSA Center	PrimaryNew_v2 (150 sources)		jwst-nirspec-g395h		1.5	
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	320	83.766251	-5.389257	23.0	1	375	83.775262	-5.443983	23.0	
	1	321	83.766348	-5.399343	23.3	1	516	83.789496	-5.449503	22.8	
	1	323	83.758621	-5.399881	23.3	1	520	83.785771	-5.439257	23.7	
	1	329	83.753871	-5.431395	22.6	1	538	83.754578	-5.393715	23.2	

Proposal 9292 - Observation 4 - How Large Stellar Flares Influence the Chemistry of Protoplanetary Disks

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G395H/F290LP)	c1 : W1P4obs4a	3 Shutter Slitlet	83.77356275 Degrees - 5.4165538888888 705 Degrees	203.16311273919 231			3	30	1610.515
	2	1 (G395H/F290LP)	c1 : W1P4obs4b	3 Shutter Slitlet	83.772567291666 66 Degrees - 5.4154802777777 6 Degrees	203.16320371216 096			3	30	1610.515
	3	1 (G395H/F290LP)	c1 : W1P4obs4c	3 Shutter Slitlet	83.781683666666 67 Degrees - 5.4194077777777 92 Degrees	203.16236013632 09			3	30	1610.515
	4	1 (G395H/F290LP)	c1 : W1P4obs4d	3 Shutter Slitlet	83.773641833333 33 Degrees - 5.4090947222222 21 Degrees	203.16309186423 425			3	30	1610.515
	5	1 (G395H/F290LP)	c1 : W1P4obs4e	3 Shutter Slitlet	83.776275708333 33 Degrees - 5.4190686111111 29 Degrees	203.16286404412 77			3	30	1610.515
Special Requirements	Between Dates 02-FEB-2026:09:06:00 and 02-FEB-2026:10:12:00 MSA Scheduled Aperture PA 203.1586 to 203.1586 Degrees (V3 64.58403 to 64.58403)										

Proposal 9292 - Observation 5 - How Large Stellar Flares Influence the Chemistry of Protoplanetary Disks

Observation	Proposal 9292, Observation 5: W2P1_Orion_Nebula										Mon Dec 01 22:01:19 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Comments: The proposed Chandra/JWST timeline is depicted in Figure 4 of the proposal. All observing dates (orange tick marks) are currently fixed in the Chandra CPS and JWST APT forms, starting from February 1, 2026, 00:00:00 UT. However, the Chandra and JWST planning teams may shift this timeline within the January-March, 2026 feasibility window to resolve any proposal date conflicts.										
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	OrionNebula_v2	RA: 05 35 17.1187 (83.8213279d) Dec: -05 22 49.41 (-5.38039d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	Filter: F140X; Readout: NRSRAPID; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold
	MSATA	false		No	MSA Center	PrimaryNew_v2 (150 sources)		jwst-nirspec-g395h		1.5	
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	154	83.854450	-5.296431	22.0	1	167	83.792927	-5.296140	20.9	
	1	156	83.842111	-5.316007	21.0	1	170	83.804850	-5.271120	22.1	
	1	163	83.839405	-5.286343	21.4	1	172	83.801769	-5.279703	22.2	
	1	164	83.810479	-5.313986	21.3	1	173	83.800904	-5.279321	21.9	

Proposal 9292 - Observation 5 - How Large Stellar Flares Influence the Chemistry of Protoplanetary Disks

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G395H/F290LP)	c1 : W2P1obs5a	3 Shutter Slitlet	83.824159749999 99 Degrees - 5.297370555555 74 Degrees	204.88082117864 45			3	30	1610.515
	2	1 (G395H/F290LP)	c1 : W2P1obs5b	3 Shutter Slitlet	83.822611875 Degrees - 5.292078055555 78 Degrees	204.88095169349 123			3	30	1610.515
	3	1 (G395H/F290LP)	c1 : W2P1obs5c	3 Shutter Slitlet	83.827611 Degrees - 5.293360833333 38 Degrees	204.88049159027 452			3	30	1610.515
	4	1 (G395H/F290LP)	c1 : W2P1obs5d	3 Shutter Slitlet	83.821791458333 34 Degrees - 5.287963611111 38 Degrees	204.88101752449 904			3	30	1610.515
	5	1 (G395H/F290LP)	c1 : W2P1obs5e	3 Shutter Slitlet	83.825735791666 66 Degrees - 5.286493333333 4 Degrees	204.88064837770 807			3	30	1610.515
Special Requirements	Between Dates 04-FEB-2026:12:00:00 and 04-FEB-2026:13:06:00 MSA Scheduled Aperture PA 204.8813 to 204.8813 Degrees (V3 66.306694 to 66.306694)										

Proposal 9292 - Observation 6 - How Large Stellar Flares Influence the Chemistry of Protoplanetary Disks

Observation	Proposal 9292, Observation 6: W2P2_Orion_Nebula										Mon Dec 01 22:01:19 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Comments: The proposed Chandra/JWST timeline is depicted in Figure 4 of the proposal. All observing dates (orange tick marks) are currently fixed in the Chandra CPS and JWST APT forms, starting from February 1, 2026, 00:00:00 UT. However, the Chandra and JWST planning teams may shift this timeline within the January-March, 2026 feasibility window to resolve any proposal date conflicts.										
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous	
	(2)	OrionNebula_v2	RA: 05 35 17.1187 (83.8213279d) Dec: -05 22 49.41 (-5.38039d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	Filter: F140X; Readout: NRSRAPID; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold
	MSATA	false		No	MSA Center	PrimaryNew_v2 (150 sources)		jwst-nirspec-g395h		1.5	
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	272	83.845070	-5.338721	21.6	1	295	83.803747	-5.373326	21.9	
	1	281	83.837123	-5.348677	21.3	1	342	83.791264	-5.356247	21.4	
	1	287	83.820396	-5.364511	21.8	1	488	83.805184	-5.369404	21.6	
	1	291	83.811259	-5.338059	21.1	1	490	83.804377	-5.359514	22.2	

Proposal 9292 - Observation 6 - How Large Stellar Flares Influence the Chemistry of Protoplanetary Disks

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G395H/F290LP)	c1 : W2P2obs6a	3 Shutter Slitlet	83.819295 Degrees - 5.3565447222222 25 Degrees	205.52616120145 936			3	30	1610.515
	2	1 (G395H/F290LP)	c1 : W2P2obs6b	3 Shutter Slitlet	83.818568083333 32 Degrees - 5.3450713888888 69 Degrees	205.52620459284 02			3	30	1610.515
	3	1 (G395H/F290LP)	c1 : W2P2obs6c	3 Shutter Slitlet	83.815854125 Degrees - 5.3577561111110 87 Degrees	205.52648343024 33			3	30	1610.515
	4	1 (G395H/F290LP)	c1 : W2P2obs6d	3 Shutter Slitlet	83.821307999999 99 Degrees - 5.3574402777777 55 Degrees	205.52597603530 185			3	30	1610.515
	5	1 (G395H/F290LP)	c1 : W2P2obs6e	3 Shutter Slitlet	83.814926916666 66 Degrees - 5.3452177777778 05 Degrees	205.52654314464 317			3	30	1610.515
Special Requirements	Between Dates 04-FEB-2026:21:00:00 and 04-FEB-2026:22:06:00 MSA Scheduled Aperture PA 205.5260 to 205.5260 Degrees (V3 66.95145 to 66.95145)										

Proposal 9292 - Observation 7 - How Large Stellar Flares Influence the Chemistry of Protoplanetary Disks

Observation	Proposal 9292, Observation 7: W2P3_Orion_Nebula										Mon Dec 01 22:01:19 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	OrionNebula_v2	RA: 05 35 17.1187 (83.8213279d) Dec: -05 22 49.41 (-5.38039d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	Filter: F140X; Readout: NRSRAPID; 8 sources in 4 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold
	MSATA	false		No	MSA Center	PrimaryNew_v2 (150 sources)		jwst-nirspec-g395h		1.5	
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	225	83.877095	-5.389515	21.8	1	253	83.863076	-5.430533	21.2	
	1	235	83.869966	-5.388101	21.9	1	269	83.847776	-5.391330	21.9	
	1	248	83.865289	-5.393540	22.2	1	351	83.826307	-5.399939	21.0	
	1	249	83.866221	-5.404223	21.5	1	463	83.833459	-5.422882	21.1	

Proposal 9292 - Observation 7 - How Large Stellar Flares Influence the Chemistry of Protoplanetary Disks

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G395H/F290LP)	c1 : W2P3obs7a	3 Shutter Slitlet	83.853701333333 33 Degrees - 5.4057052777777 7 Degrees	205.47069381322 672			3	30	1610.515
	2	1 (G395H/F290LP)	c1 : W2P3obs7b	3 Shutter Slitlet	83.850774249999 99 Degrees - 5.4092708333333 235 Degrees	205.47097145939 875			3	30	1610.515
	3	1 (G395H/F290LP)	c1 : W2P3obs7c	3 Shutter Slitlet	83.861546166666 66 Degrees - 5.4051027777777 56 Degrees	205.46996484218 68			3	30	1610.515
	4	1 (G395H/F290LP)	c1 : W2P3obs7d	3 Shutter Slitlet	83.858653458333 34 Degrees - 5.4114386111111 23 Degrees	205.47024358686 27			3	30	1610.515
	5	1 (G395H/F290LP)	c1 : W2P3obs7e	3 Shutter Slitlet	83.851165958333 34 Degrees - 5.4026719444444 21 Degrees	205.47092407353 503			3	30	1610.515
Special Requirements	Between Dates 05-FEB-2026:06:00:00 and 05-FEB-2026:07:06:00 MSA Scheduled Aperture PA 205.4737 to 205.4737 Degrees (V3 66.899086 to 66.899086)										

Proposal 9292 - Observation 8 - How Large Stellar Flares Influence the Chemistry of Protoplanetary Disks

Observation	Proposal 9292, Observation 8: W2P4_Orion_Nebula										Mon Dec 01 22:01:19 GMT 2025
	Diagnostic Status: Warning										
Diagnostics	Observing Template: NIRSpec MultiObject Spectroscopy										
	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	OrionNebula_v2	RA: 05 35 17.1187 (83.8213279d) Dec: -05 22 49.41 (-5.38039d) Equinox: J2000								
Acquisition	Comments: Description=[]										
	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	Filter: F140X; Readout: NRSRAPID; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method		HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map	Spectral Overlap Threshold
	MSATA		false		No	MSA Center	PrimaryNew_v2 (150 sources)		jwst-nirspec-g395h		1.5
Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	298	83.802282	-5.415054	21.8	1	357	83.807522	-5.396420	21.0	
	1	305	83.783826	-5.386814	21.4	1	359	83.795902	-5.396753	21.5	
	1	311	83.777748	-5.436614	21.8	1	483	83.808477	-5.395045	22.0	
	1	354	83.812805	-5.406116	21.1	1	485	83.807515	-5.394475	21.8	

Proposal 9292 - Observation 8 - How Large Stellar Flares Influence the Chemistry of Protoplanetary Disks

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G395H/F290LP)	c1 : W2P4obs8a	3 Shutter Slitlet	83.782103083333 33 Degrees - 5.4134063888889 16 Degrees	205.43392374834 175			3	30	1610.515
	2	1 (G395H/F290LP)	c1 : W2P4obs8b	3 Shutter Slitlet	83.772836083333 33 Degrees - 5.4165399999999 98 Degrees	205.43479423396 977			3	30	1610.515
	3	1 (G395H/F290LP)	c1 : W2P4obs8c	3 Shutter Slitlet	83.771301041666 66 Degrees - 5.4184277777777 65 Degrees	205.43494096275 975			3	30	1610.515
	4	1 (G395H/F290LP)	c1 : W2P4obs8d	3 Shutter Slitlet	83.782832416666 66 Degrees - 5.4133669444444 195 Degrees	205.43385564012 343			3	30	1610.515
	5	1 (G395H/F290LP)	c1 : W2P4obs8e	3 Shutter Slitlet	83.776771791666 66 Degrees - 5.4192641666666 646 Degrees	205.43443174748 86			3	30	1610.515
Special Requirements	Between Dates 05-FEB-2026:12:06:00 and 05-FEB-2026:13:12:00 MSA Scheduled Aperture PA 205.4302 to 205.4302 Degrees (V3 66.85564 to 66.85564)										